YEAR 10 MATHS · CHAPTER 4 · MIND MAP

Measurement and surds

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1 Irrational numbers and surds

A surd is a root that isn't a whole number, like $\sqrt{2}$. Simplify using square factors.

$\sqrt{8} = 2\sqrt{2}$ $\sqrt{48} = \sqrt{16}\sqrt{3} = 4\sqrt{3}$

2 Operations with surds

Add and subtract like surds only. Multiply or divide the numbers outside and inside separately.

$\sqrt{12} + \sqrt{27} = 2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$

$\sqrt{6} \times \sqrt{3} = \sqrt{18} = 3\sqrt{2}$ $6\sqrt{10} \div 2\sqrt{5} = 3\sqrt{2}$

3 Rationalising the denominator

Multiply top and bottom so there's no surd on the bottom.

$$\frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

$$\frac{1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = \frac{\sqrt{3}-1}{2}$$

4 Review of length

Circumference $2\pi r$. Arc length $\frac{\theta}{360} \times 2\pi r$.

$r = 10,\ \theta = 72°$: arc $\frac{1}{5} \times 20\pi = 4\pi \approx 12.57$

5 Pythagoras in 3D

Find a right triangle, or use $d = \sqrt{a^2 + b^2 + c^2}$ for a box.

Box 2 by 3 by 6: $\sqrt{4 + 9 + 36} = 7$

6 Review of area

Triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2, sector $\frac{\theta}{360}\pi r^2$

Sector $r = 6,\ \theta = 60°$: $\frac{1}{6} \times 36\pi = 6\pi \approx 18.85$

7 Measurement errors and accuracy

The error is up to half the smallest unit. Percentage error = error ÷ measurement × 100%.

4.2 m to the nearest 0.1 m: between 4.15 m and 4.25 m

Percentage error $\frac{0.05}{4.2} \times 100\% \approx 1.2\%$

8 Surface area: prisms and cylinders

Add every face. Cylinder: $2\pi r^2 + 2\pi rh$.

Cylinder $r = 4,\ h = 10$: $32\pi + 80\pi = 112\pi \approx 351.9$

9 Surface area: pyramids and cones

Cone: $\pi r^2 + \pi rs$, where s is the slant height.

$r = 3,\ h = 4$: $s = 5$, so $SA = 9\pi + 15\pi = 24\pi \approx 75.4$

Watch out: use the slant height, not the vertical height.

10 Volume: prisms and cylinders

$V = Ah$. Cylinder: $V = \pi r^2 h$.

$r = 5,\ h = 8$: $V = 200\pi \approx 628.3$

11 Volume: pyramids and cones

$V = \frac{1}{3}Ah$: a third of the matching prism.

Cone $r = 3,\ h = 4$: $\frac{1}{3} \times 9\pi \times 4 = 12\pi \approx 37.7$

Square pyramid, base 6, height 10: $\frac{1}{3} \times 36 \times 10 = 120$

12 Spheres

$SA = 4\pi r^2$ and $V = \frac{4}{3}\pi r^3$

$r = 6$: $SA = 144\pi \approx 452.4$

$V = \frac{4}{3}\pi(216) = 288\pi \approx 904.8$